



Rapid Prototyping and User-Centric ML Applications in the post LLM-era

Benjamin Aanes – Head of AI Products @ Crayon Consulting

NHH - Feb 17th 2025



KEY MESSAGES OF THIS LECTURE

Key messages of this lecture

- Understanding the timeline for the evolution of AI, how we got here, and how LLMs are related to statistical learning concepts
- Understanding the purpose of AI in a business setting, the interconnection between business strategy and AI Products, and the importance of user-centric development
- Understanding how the advancements in LLMs and AI tools have democratized prototyping and accelerated product development, and how to utilize these tools to build your own AI Products

INTRO

Who am I?

BearingPoint®

*Data Science
Consultant*

unacast.

*Senior Data Scientist /
Team Lead*

 **Crayon**
CONSULTING

Head of AI Products

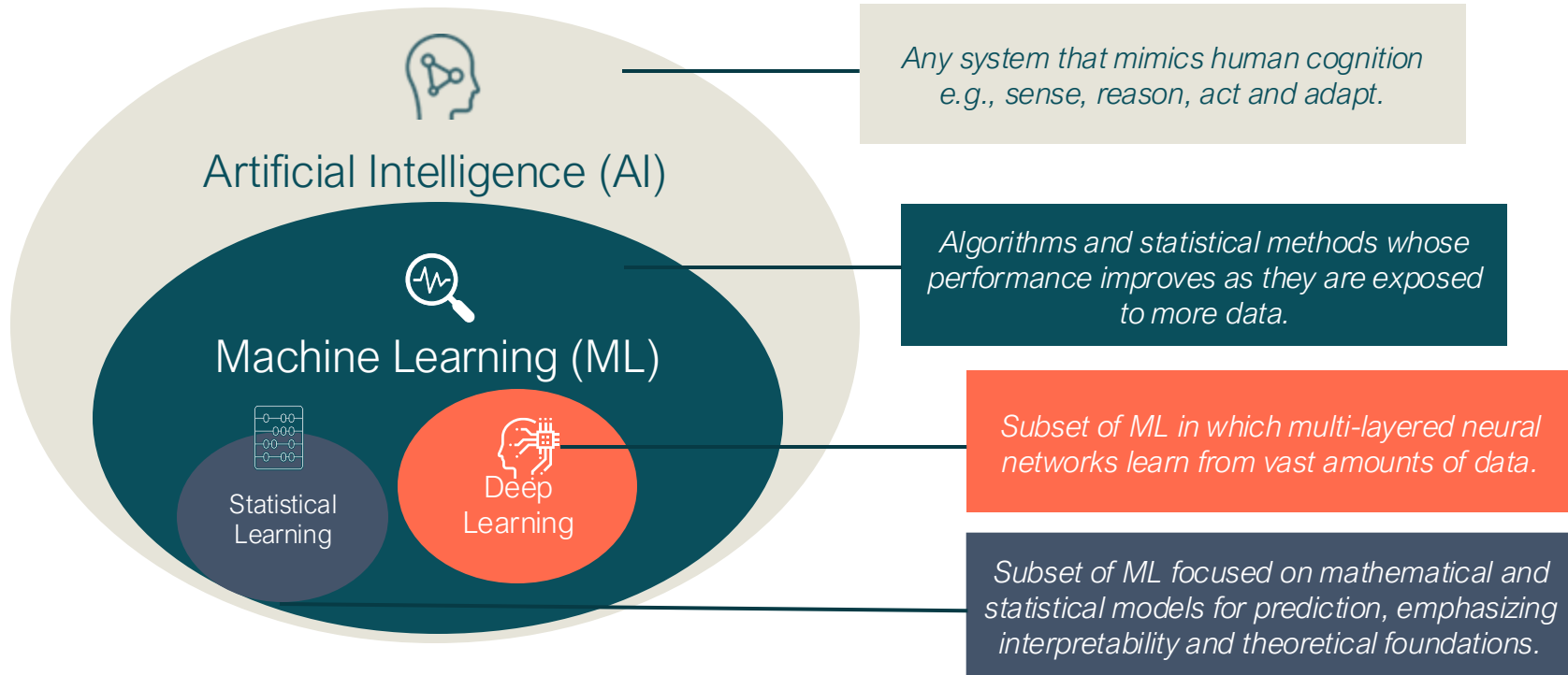
NHH 

*Finance / Business
Analytics*

AI: How did we get
here?

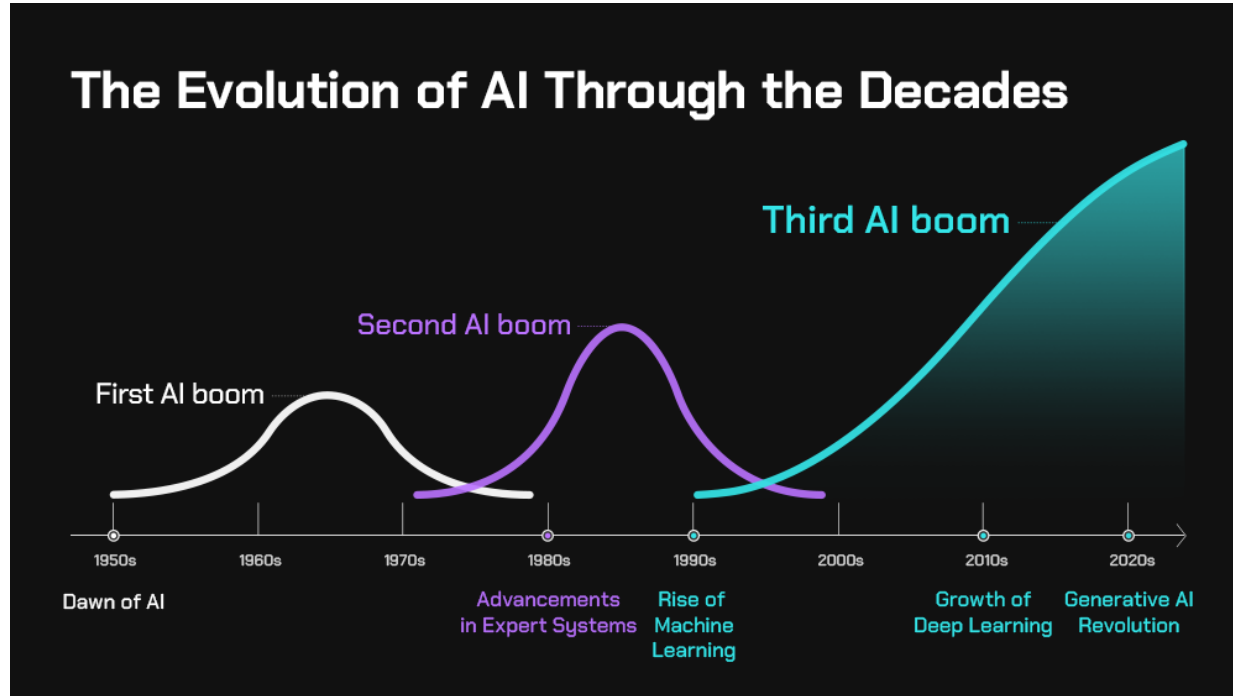
QUICKLY ON TERMINOLOGY

Not too long ago, you could tell if somebody was technical or not based on their choice between saying «AI» or «ML» – This is no longer the case



FROM STATISTICAL LEARNING TO BILLION DOLLAR LARGE LANGUAGE MODELS

The evolution of AI is characterized by cycles of hype and disappointment. Understanding key factors can help us understand if another winter will come

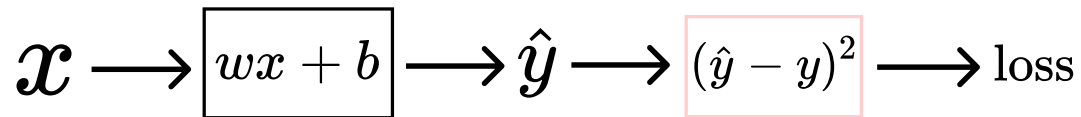


[Borrowed from skhvnix](#)

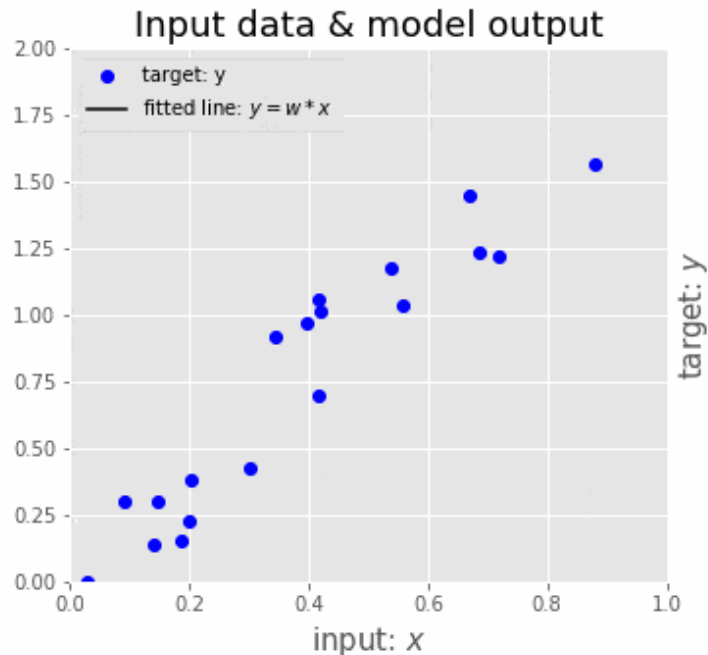
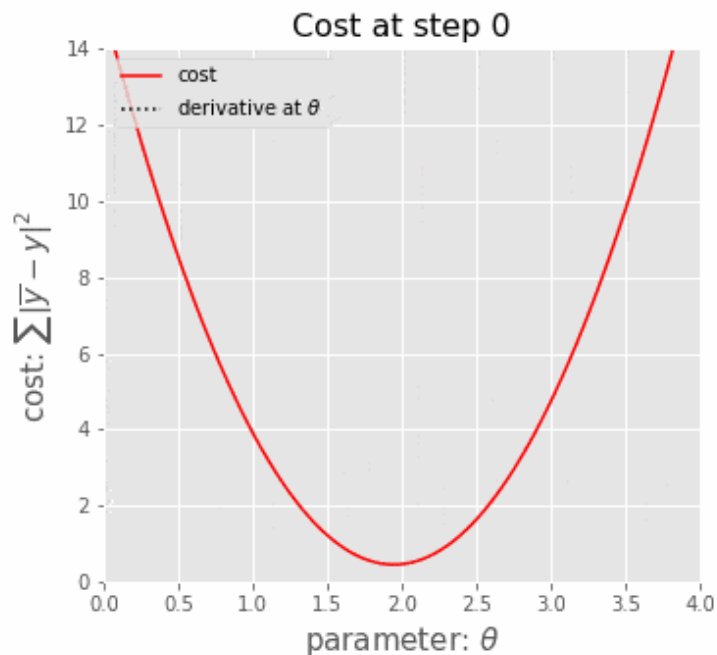
KEY BUILDING BLOCKS OF MODERN AI: DATA, MODEL, LOSS

ML uses data, labels, models, and loss functions to learn patterns, minimizing errors to optimize predictions or decisions

- Input data and labels x, y
- Model, for instance linear regression: $y = wx + b$
- Loss function, for instance squared error: $(\hat{y} - y)^2$
- Goal: Adjust w and b to minimize the loss function

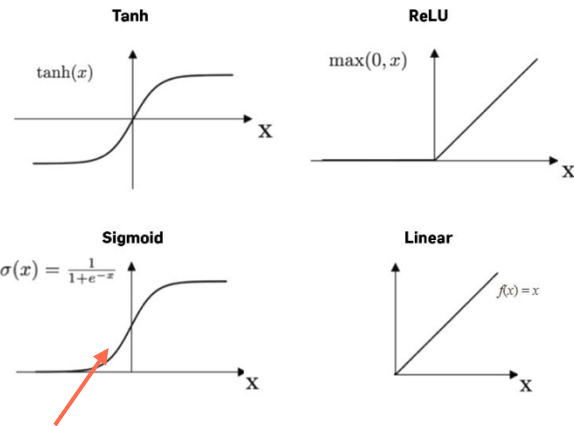
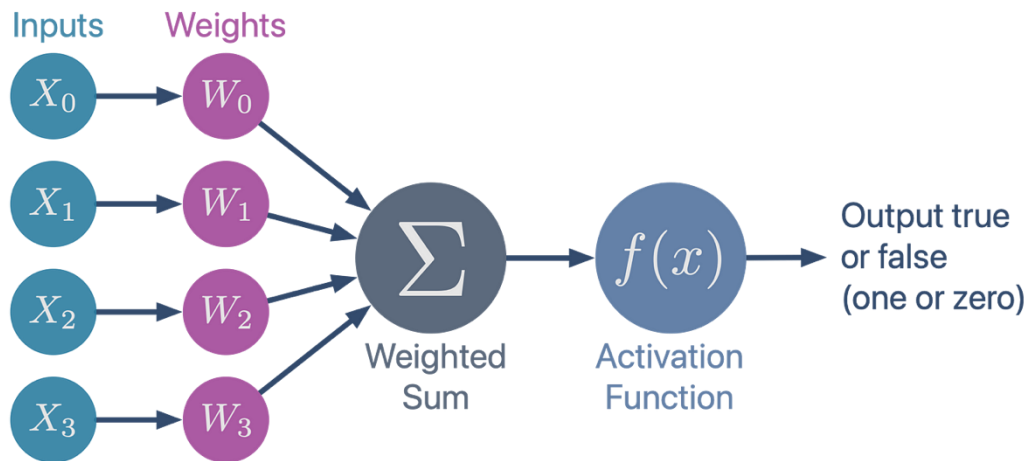


Model parameters are updated iteratively through gradient descent and backpropagation, minimizing errors by adjusting weights based on feedback



COMPUTATIONAL NEURON

Extending linear regression, adding a non-linear activation function, we end up with a computational neuron – the start of neural networks

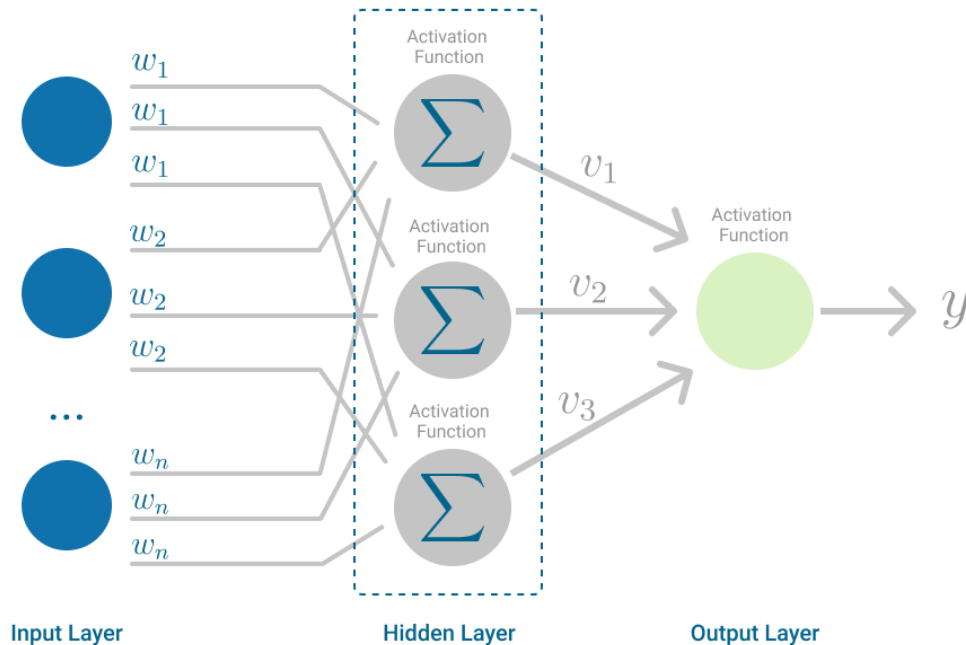


Logistic Regression

[Borrowed from aimind.so](https://aimind.so)

MULTILAYER PERCEPTRON

By stacking neurons in dense structures we can define architectures that are universal function approximators – they can approximate any function



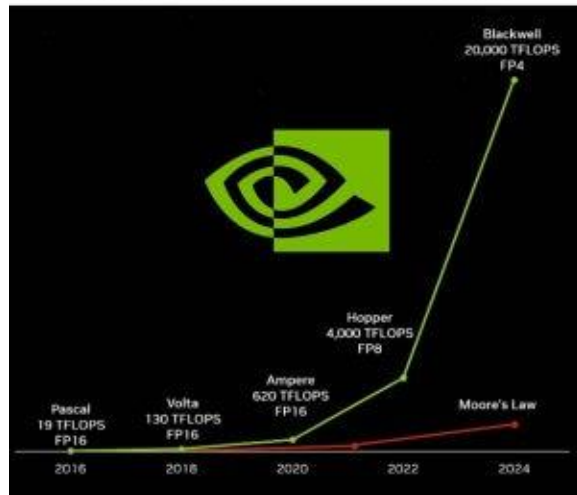
[Borrowed from Muhammad Rafly Indrawan @ Medium](#)

DEEP LEARNING NECESSITATES GPU TRAINING

Deep learning was for a long time considered theoretically interesting but infeasible due to the vast computational requirements

- Every computation within a multilayer perceptron can be performed with a matrix multiplication (matmuls)
- What hardware do we already have that are optimized for parallelization of matmuls? → Graphics Cards
- GPU accelerated training is necessary for deep learning to work
- A key reason for NVIDIA's dominance: CUDA integration into deep learning frameworks

NVIDIA, with no solid competition, is out here competing against **Moore's Law** instead.

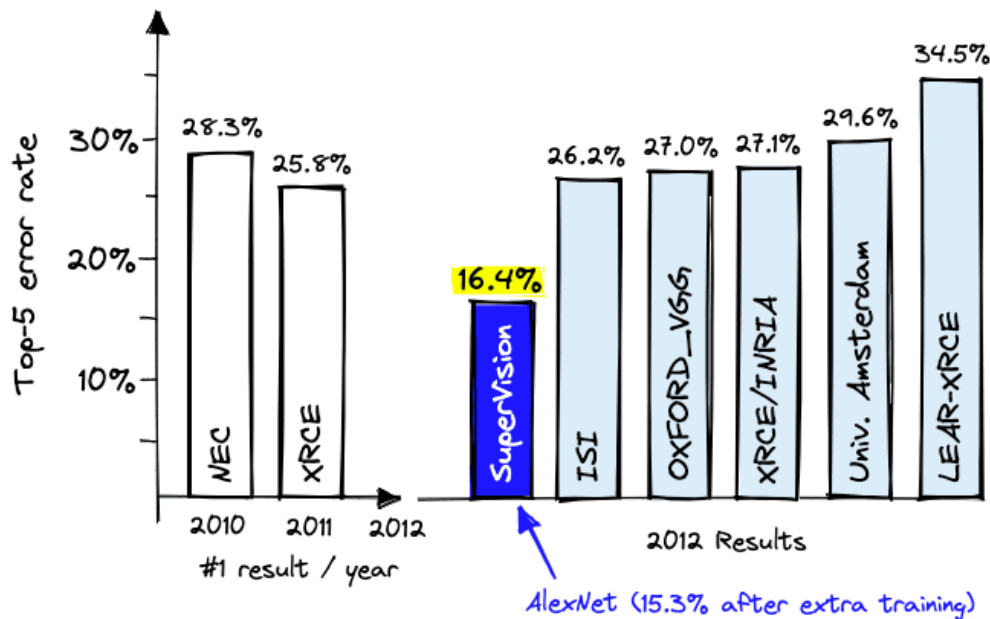


*This visual is a bit sneaky from Nvidia's side. Notice the change in floating point bits from FP16, to FP8, to FP4

[Borrowed from Linas Beliusas @ LinkedIn](#)

ALEXNET

The real birth of Deep Learning as a serious field of research was when AlexNet won the ImageNet competition with a landslide in 2012

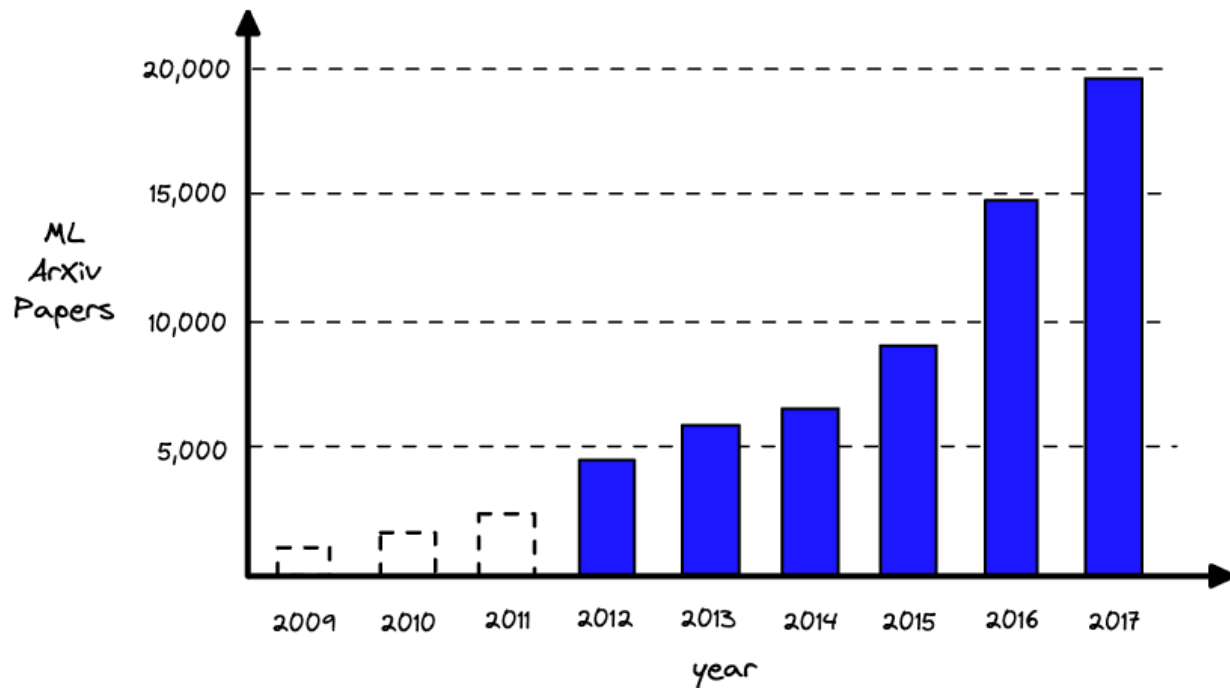


- AlexNet proved deep learning could outperform traditional methods by a wide margin, thanks to advances in architecture, activation functions, and GPU acceleration.
- Trained using 2x NVIDIA GTX 580 GPUs across 5 to 6 days on 1.2 million images

[Borrowed from Pinecode - Image Search Series](#)

MORE RESEARCH

AlexNet's success in 2012 proved deep learning's potential, leading to exponential growth in research publications

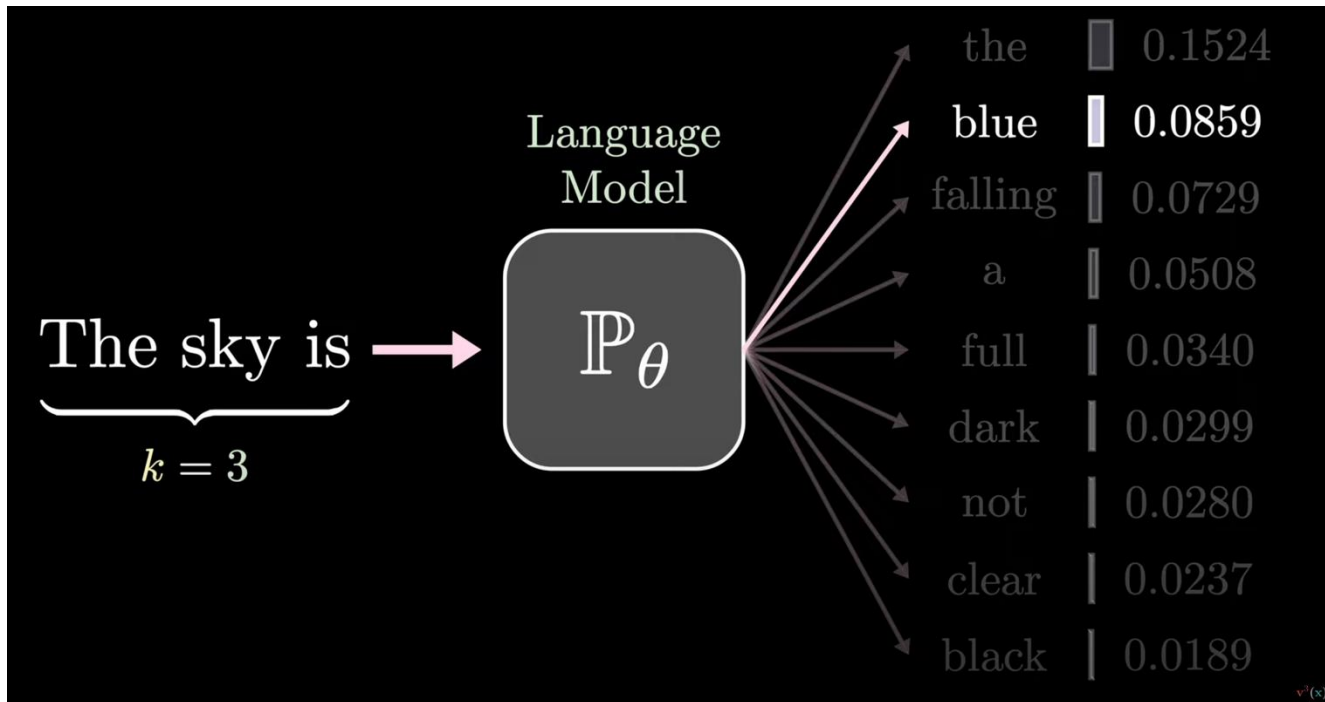


[Borrowed from Pinecone - Image Search Series](#)

<https://www.pinecone.io/learn/series/image-search/imagenet/>

LANGUAGE MODELLING

While AlexNet advanced image classification, researchers also explored how neural networks could model and generate language



[Borrowed from vcubingx @ Youtube \(Recommended Watch\)](#)

TRANSFORMERS

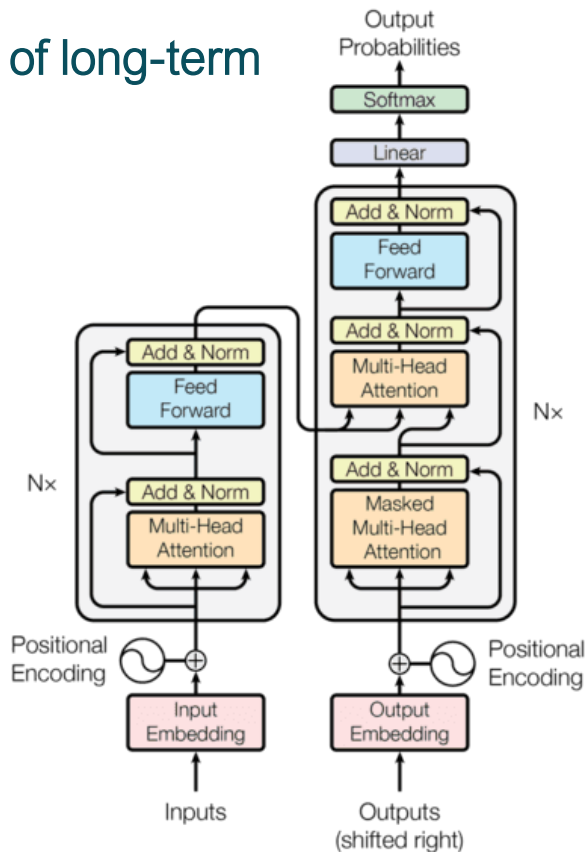
The transformer architecture solved the problem of long-term dependence in a GPU-scalable way

The *Transformers* model solves this by:

- Using *self-attention* to capture long-range dependencies efficiently
- Encoding words as *embeddings* that retain contextual meaning
- Processes *entire sequences in parallel*, making it highly scalable on GPUs

Key resources for learning more about neural networks and transformers:

- [3Blue1Brown Neural Networks Series](#)
- [The Illustrated Transformer](#)



[Attention is all you need](#)

KEY TRANSFORMER CONCEPTS

Transformers rely on a few key ideas to understand and generate language efficiently

1. Embeddings: Words can be encoded as vectors of real numbers that capture meaning and context. Adding or subtracting vectors blends meanings by imprinting one concept onto another. $E(\text{pizza}) - E(\text{Italy}) + E(\text{Germany}) \approx E(\text{Bratwurst})$
2. Semantic Similarity: Words with similar meanings are closer in this vector space
3. Attention: Transformers use self-attention to weigh the importance of all surrounding words, helping understand context dynamically.
4. Generating Text: Once trained, a Transformer generates text by predicting and appending words iteratively, forming coherent sentences.

Previously suggested resources gives a better explanation of why these facts are true than I ever could

[3blue1brown - Visualizing Transformers Talk](#)

GPT3 DATASET

What if we trained multi-billion-parameter transformer models on the entire internet on thousands of GPUs?

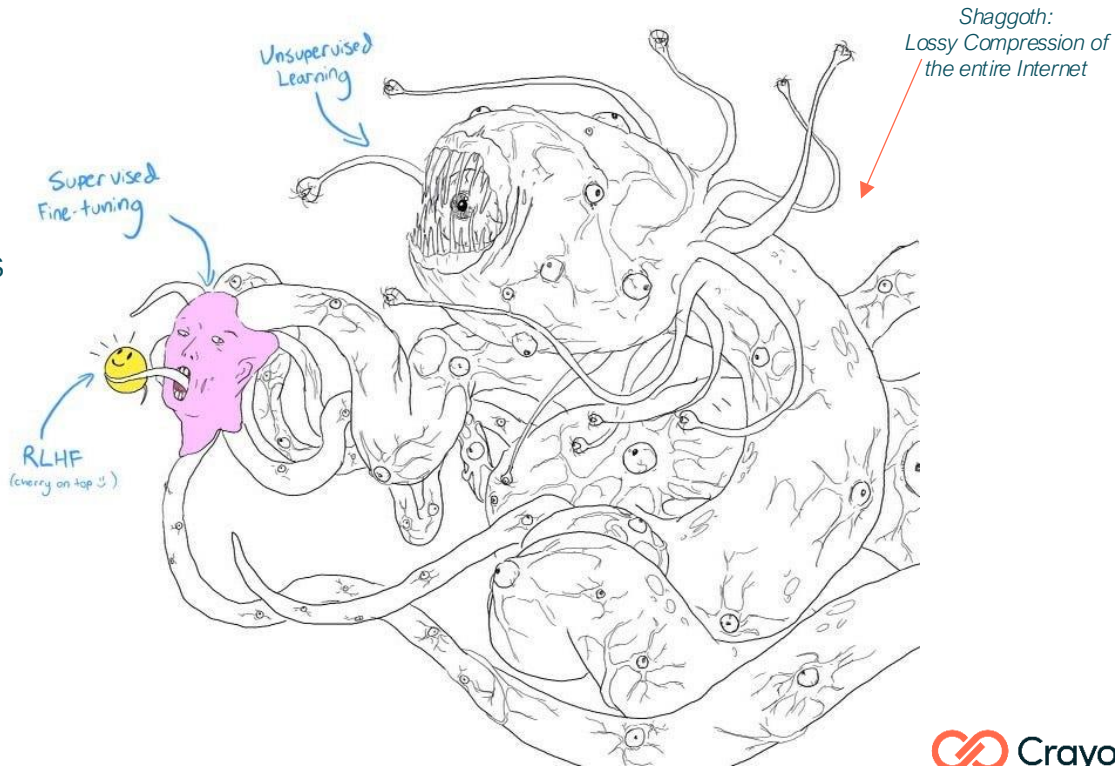
<i>Dataset</i>	Quantity (tokens)	Weight in training mix
<i>Common Crawl (filtered)</i>	410 billion	60%
<i>WebText2</i>	19 billion	22%
<i>Books1</i>	12 billion	8%
<i>Books2</i>	55 billion	8%
<i>Wikipedia</i>	3 billion	3%

[Borrowed from Pinecone - Transformer Series](#)

THREE-STAGE TRAINING PARADIGM

Large language models like GPT3/GPT4-o go through a three-step training process, aligning the models outputs to human needs

1. Unsupervised Learning: Train on next-word prediction on huge dataset
2. Supervised fine-tuning: Train the model further on curated datasets with specific instructions or tasks
3. Reinforcement learning with human feedback (RLHF): Refine model outputs by incorporating human feedback



[Borrowed from lesswrong.com](https://lesswrong.com)

FINDINGS FROM AI RESEARCHERS

At this type of scale, the task of predicting the next word is massively multi-task learning

Example tasks from next-word prediction

<u>Task</u>	<u>Example sentence in pre-training that would teach that task</u>
<i>Grammar</i>	In my free time, I like to {code, banana}
<i>Lexical semantics</i>	I went to the store to buy papaya, dragon fruit, and {durian, squirrel}
<i>World knowledge</i>	The capital of Azerbaijan is {Baku, London}
<i>Sentiment analysis</i>	Movie review: I was engaged and on the edge of my seat the whole time. The movie was {good, bad}
<i>Translation</i>	The word for “pretty” in Spanish is {bonita, hola}
<i>Spatial reasoning</i>	Iroh went into the kitchen to make tea. Standing next to Iroh, Zuko pondered his destiny. Zuko left the {kitchen, store}
<i>Math question</i>	Arithmetic exam answer key: $3 + 8 + 4 = \{15, 11\}$

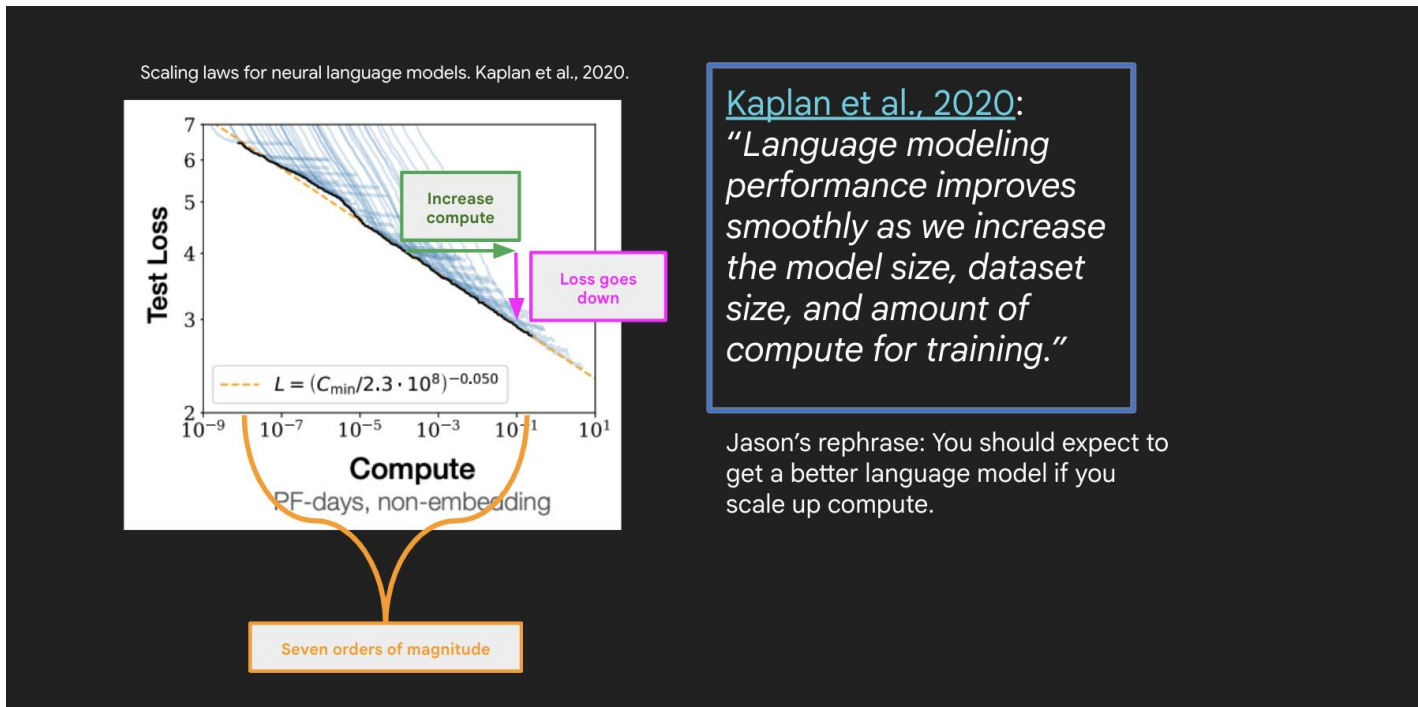
[millions more]

Extreme multi-task learning!

[Borrowed from Stanford CS25 - Guest Lecture \(Highly recommended watch\)](#)

FINDINGS FROM AI RESEARCHERS

Scaling laws show that increasing compute, model size, and data leads to predictable performance improvements, following a power-law relationship

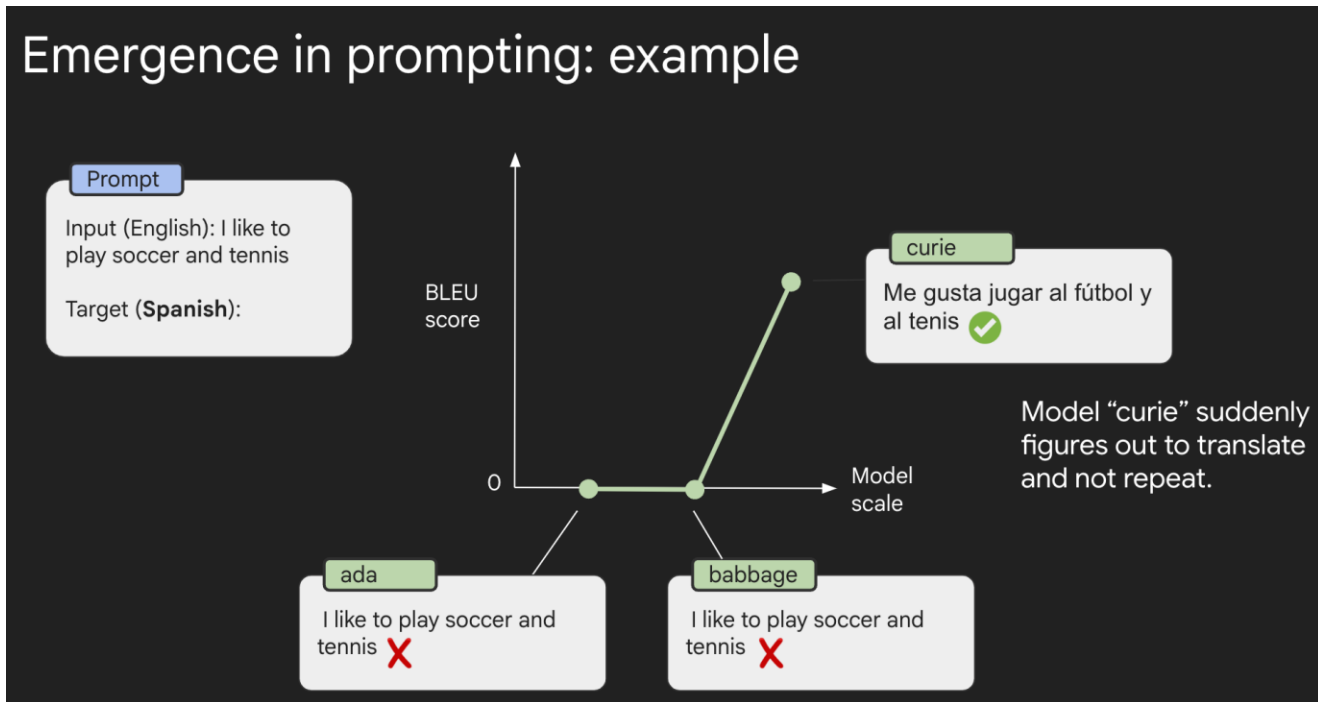


[Borrowed from Stanford CS25 - Guest Lecture \(Highly recommended watch\)](#)

[Scaling Laws for Neural Language Models, Kaplan\(2020\).](#)

FINDINGS FROM AI RESEARCHERS

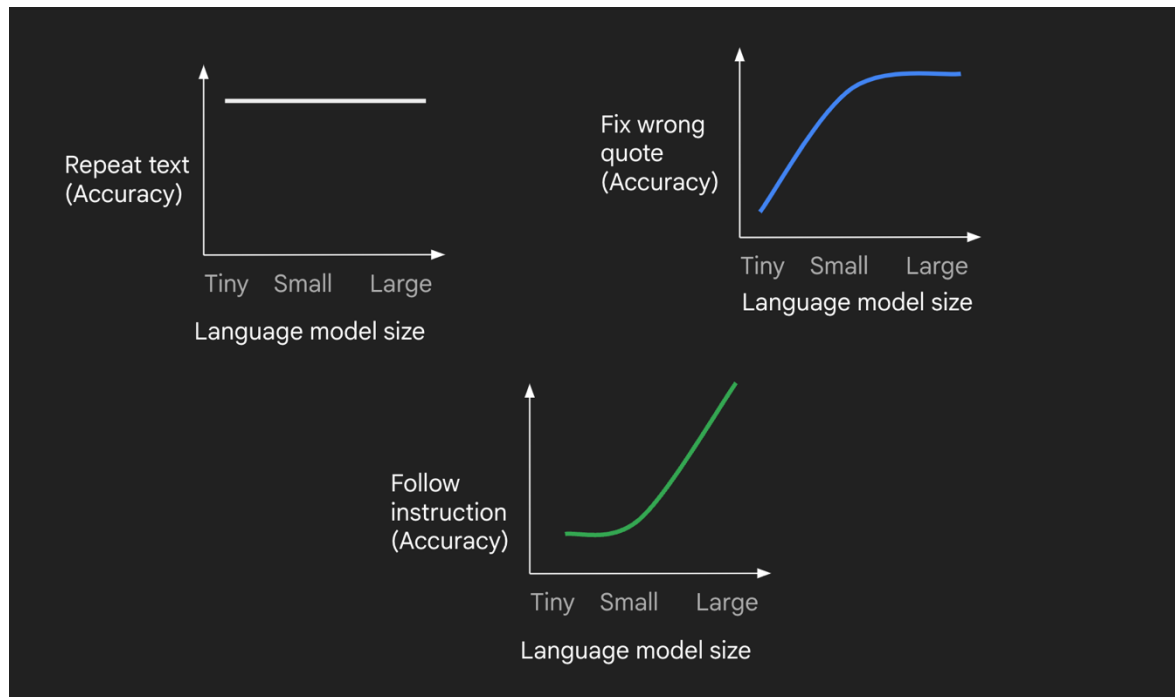
While overall performance scales smoothly, the performance on a given task may scale in an emergent fashion



[Borrowed from Stanford CS25 - Guest Lecture \(Highly recommended watch\)](#)

FINDINGS FROM AI RESEARCHERS

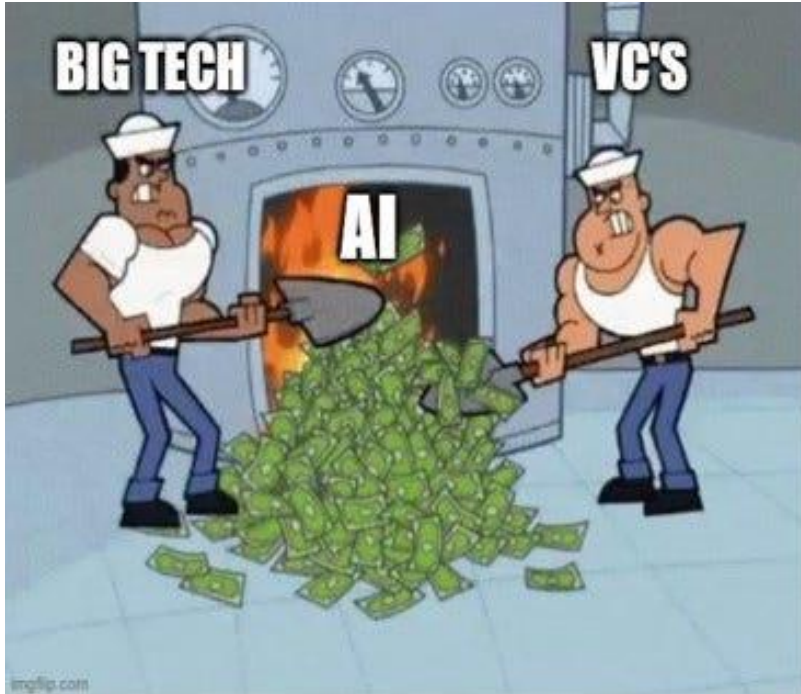
And different types of tasks may scale in different ways, for instance the model might become better at following instructions indefinitely



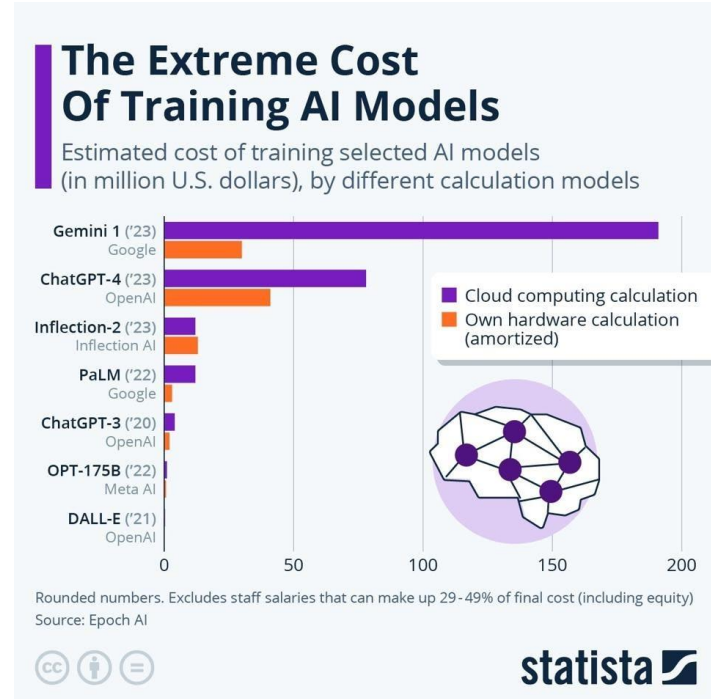
[Borrowed from Stanford CS25 - Guest Lecture \(Highly recommended watch\)](#)

RELIABLE PERFORMANCE IMPROVEMENT + HYPE = INVESTMENT

AI's Promise Fuels an Investment Gold Rush



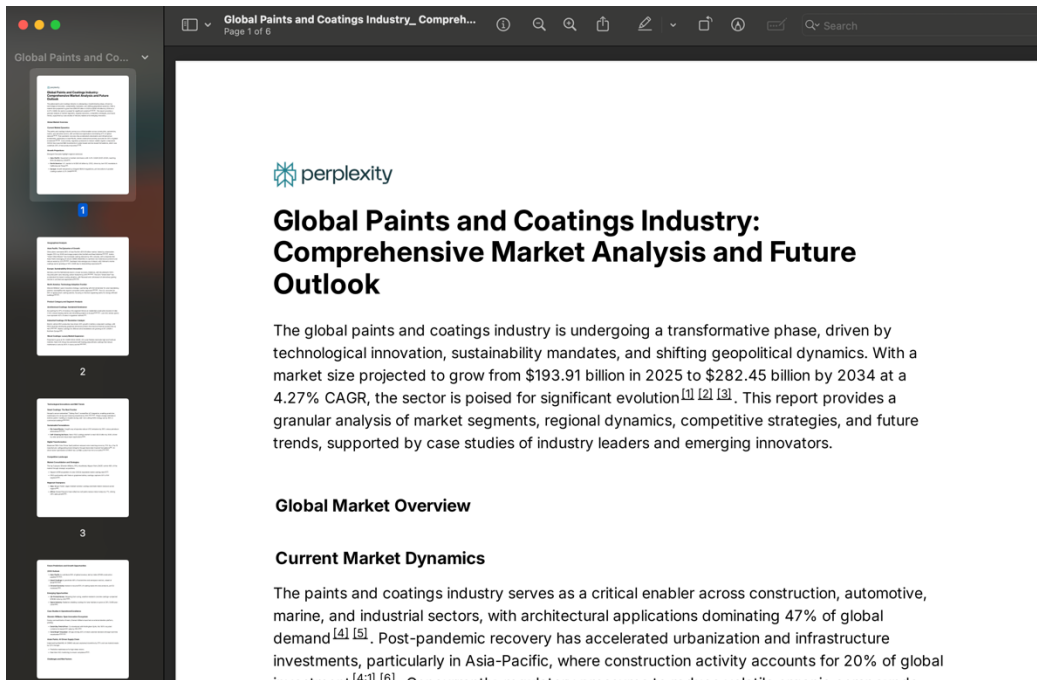
[Borrowed from Walulya Francis @Threads](#)



[Borrowed from statista.com skhvnix](#)

CURRENT STATE

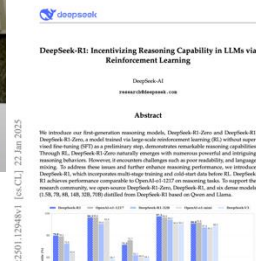
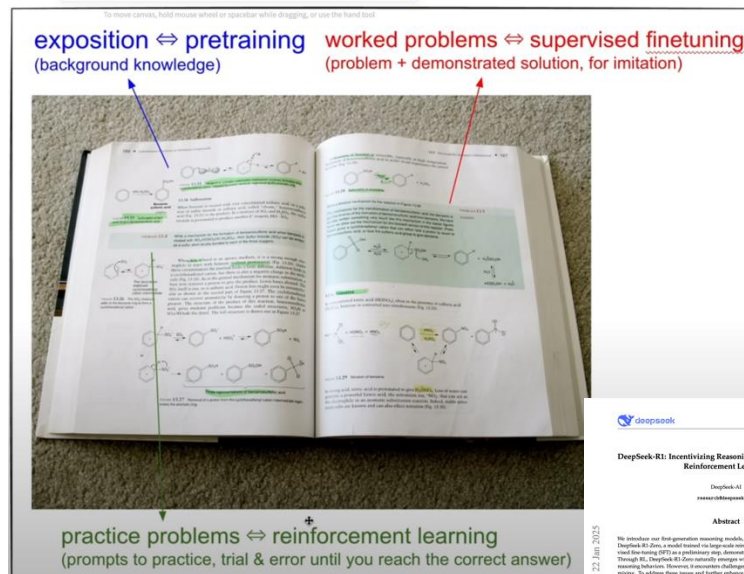
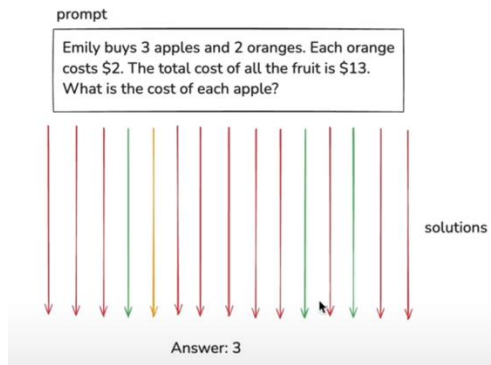
Modern AI features, such as OpenAI or Perplexity AI's deep research increasingly feel like magic



REINFORCEMENT LEARNING, REASONING

Many of these new *reasoning* features are based on a new training paradigm of reinforcement learning, such as that behind Deepseek R1

- Traditional models rely on supervised fine-tuning (SFT).
- DeepSeek R1 & OpenAI's o1, o3 use reinforcement learning (RL) instead.
- RL enables models to develop reasoning abilities without explicit human labels.

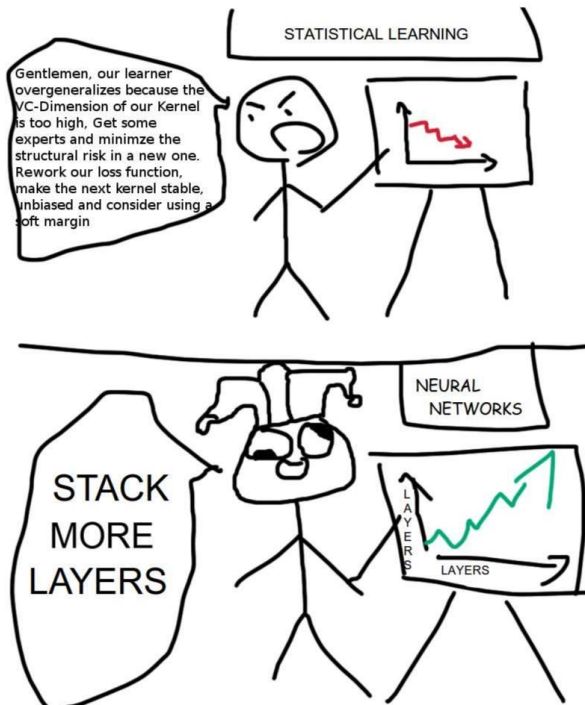


[DeepSeek-R1 Paper \(2025\)](#)

[Andrej Karpathy Deep Dive Into LLMs like ChatGPT \(Highly Recommended\)](#)

WRAP-UP ON EVOLUTION OF AI

You could argue the «stack more layers» strategy has succeeded in taking us towards true AI



[Borrowed from ProgrammerHumor @ Reddit](#)

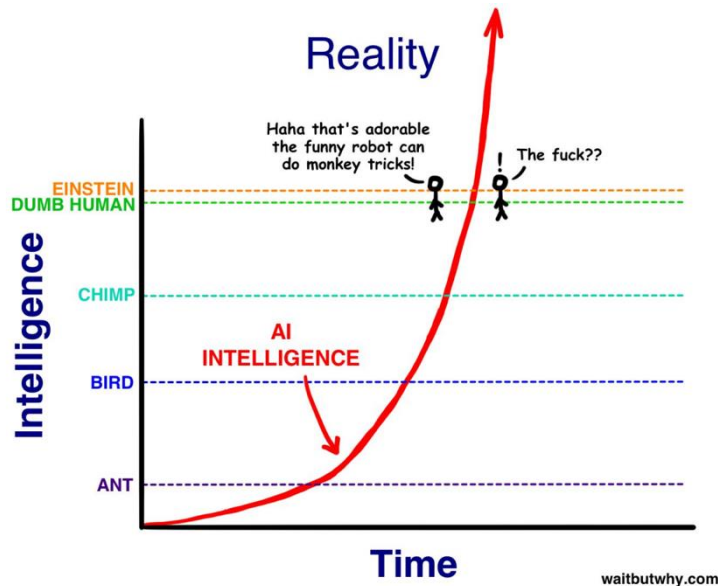
- Scaling isn't just about more layers—it's a general strategy that keeps working across different AI paradigms:
- The history of AI suggests that brute-force scaling—whether through bigger models, more compute, or more reinforcement learning trials—consistently outperforms manually engineered approaches.

[«The Bitter Lesson»](#) by Rich Sutton is an interesting reflection on this

WRAP-UP ON EVOLUTION OF AI: WHAT'S NEXT?

Will scaling laws continue indefinitely? Are we closing in on the AI Singularity?

- Are we seeing diminishing marginal returns to scaling in AI?
- Short term answer: We will see returns to smarter architectures (e.g., mixture of experts) and compute optimizations (both e.g. Deepseek) but pure brute-force scaling might be slowing down
- Long-term answer: ???

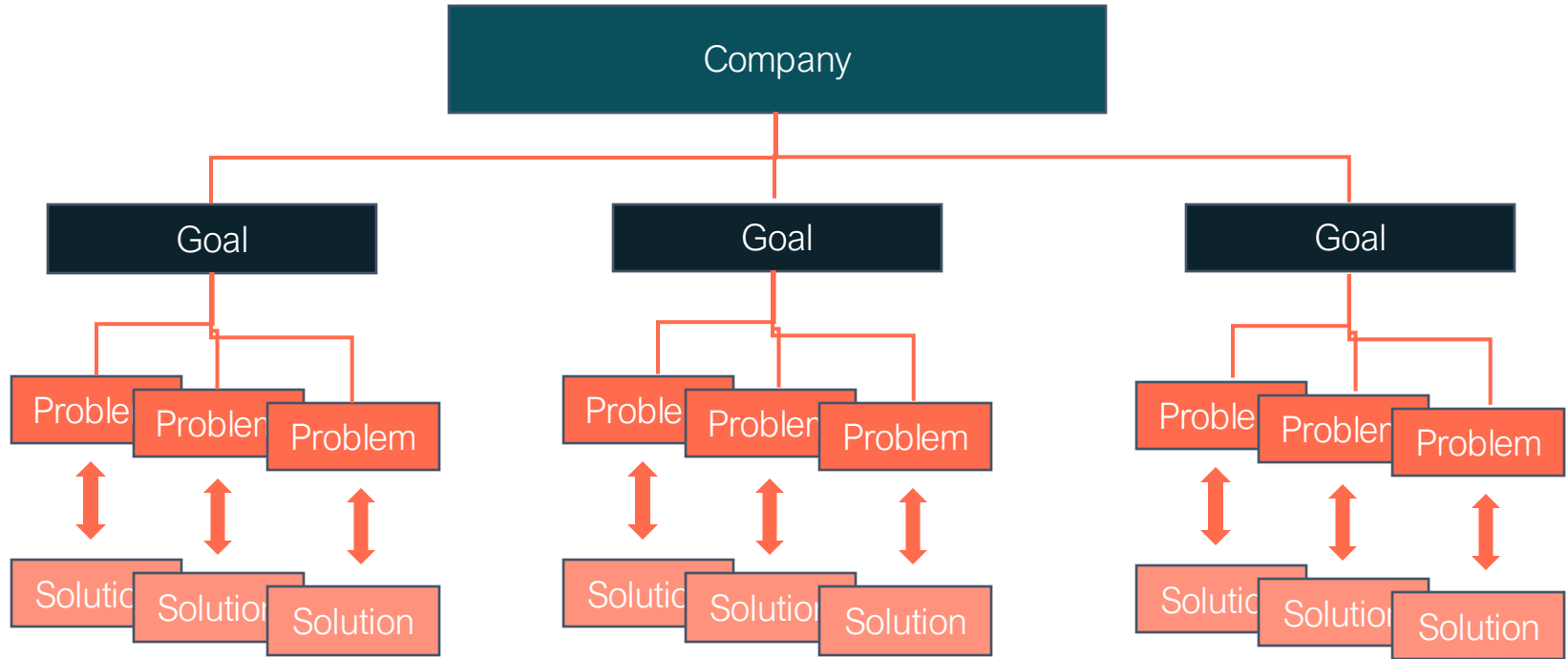


[Borrowed from ProgrammerHumor @ Reddit](#)

AI in the context of business

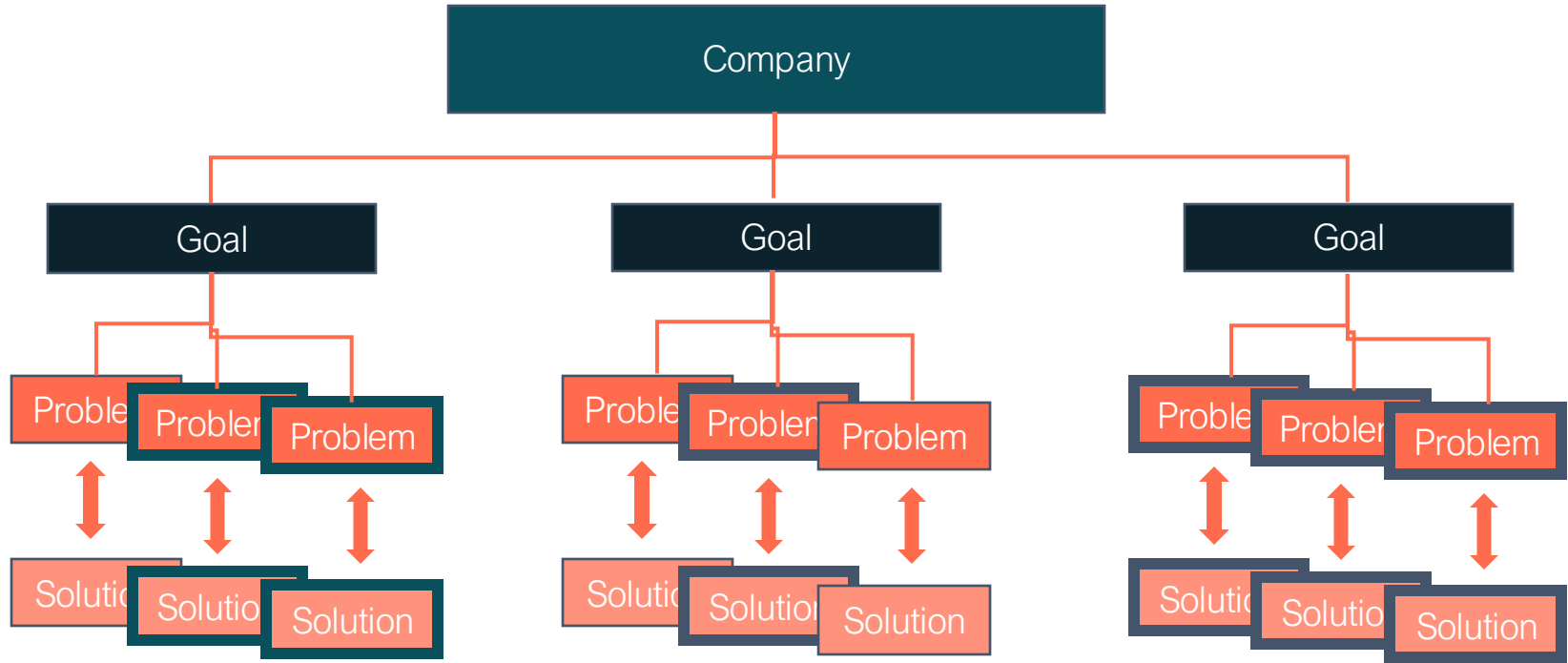
SIMPLIFIED VIEW OF BUSINESS STRATEGY

A company is a group of people working together to achieve certain objectives, solving the problems in their way



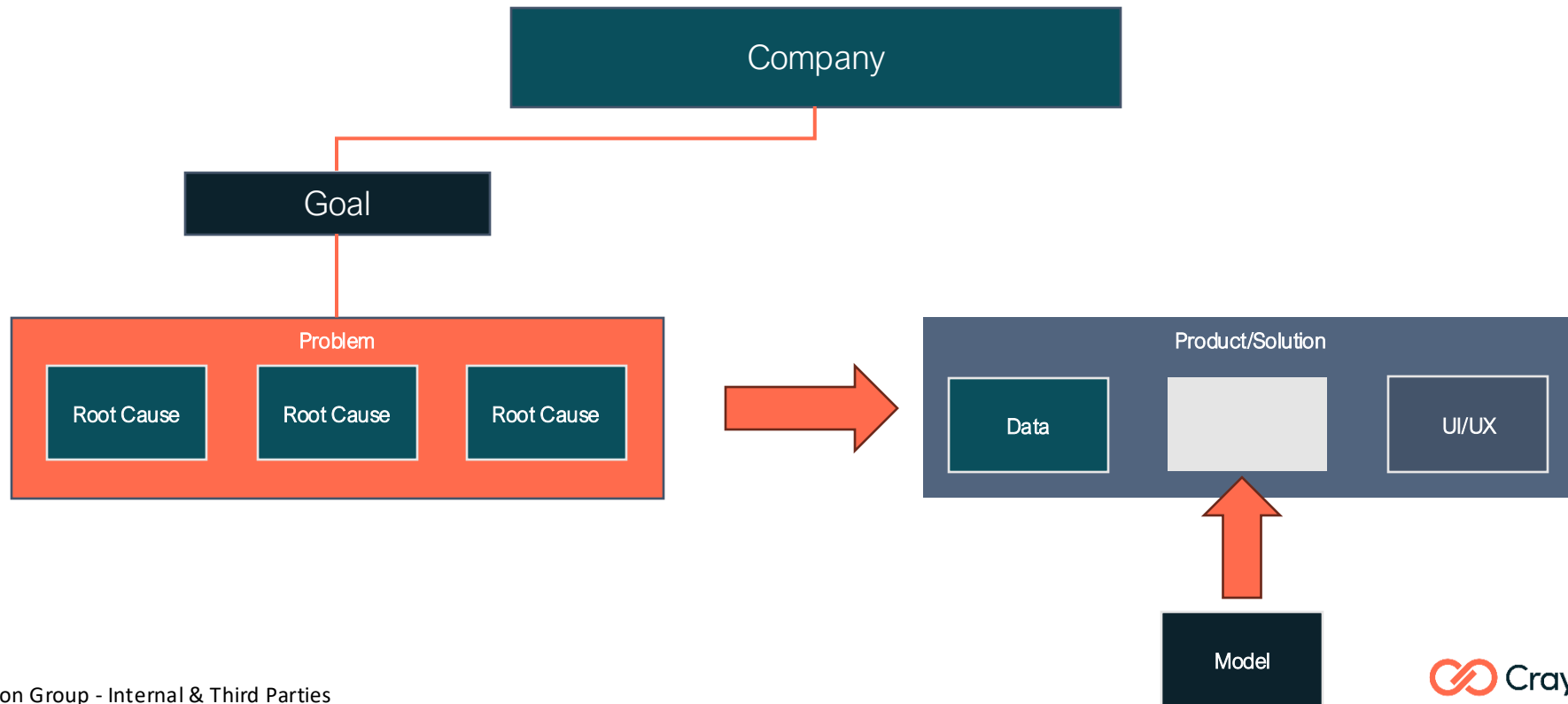
SIMPLIFIED VIEW OF AI STRATEGY

AI Strategy is about selecting the right problems to solve with AI in order to drive business impact



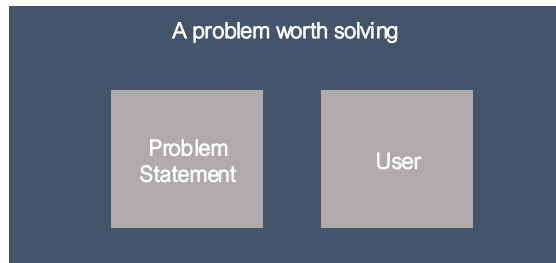
SIMPLIFIED VIEW OF AN AI PRODUCT OR SOLUTION

A good AI product combines the right data, a well-optimized model, and a user interface that drives action

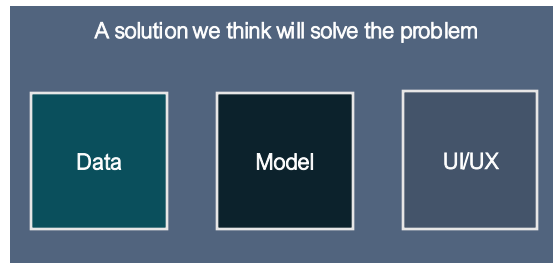


WHAT IS OUR PATH TO DELIVERING VALUE WITH AN AI PRODUCT OR SOLUTION?

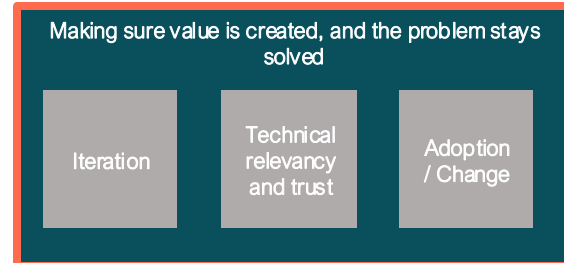
Many AI projects fail not because the model is bad, but because it doesn't integrate into workflows, isn't adopted by users, or doesn't solve the problem in a usable way



Consistently find problems worth solving



Consistently deliver solutions we are confident will solve the problem



Consistently utilize solutions to solve problems

CORE CHALLENGES WITH ML INITIATIVES

Most ML projects fail because they are treated as tech experiments rather than business solutions

Gartner: ~85% of AI projects have not delivered for their sponsors



ML is part research, therefore it is very challenging to aim for 100% success rate.

[Based on Full Stack Deep Learning Lectures, UC Berkeley, 2019/2021](#)

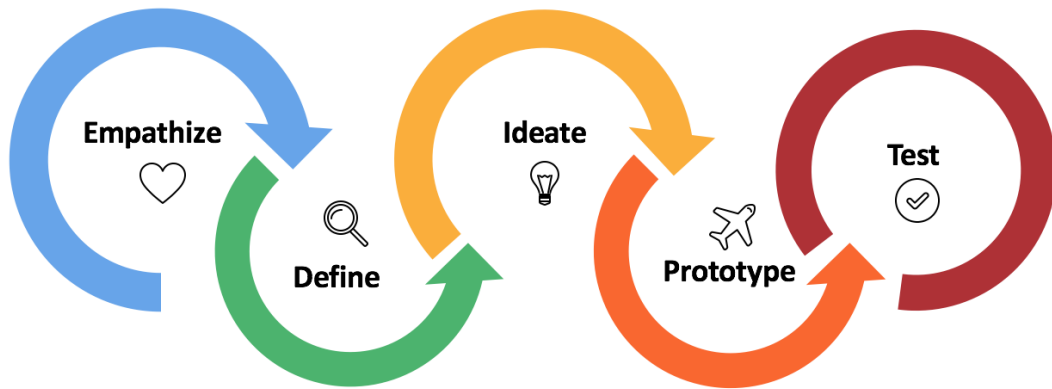
[AI, Localization Winners, Losers, Heroes, Specators and you \(2019\)](#)[Based on Full Stack Deep Learning Lectures, UC Berkeley, 2019/2021](#)

USER-CENTRIC ML DEVELOPMENT

Many ML projects fail because they optimize for technical performance instead of solving real user problems. A user-centric approach ensures AI is useful, usable, and adopted

Show don't tell: Make sure we build the right thing for the right people

Fail Fast: Make sure we find core problems sooner rather than later

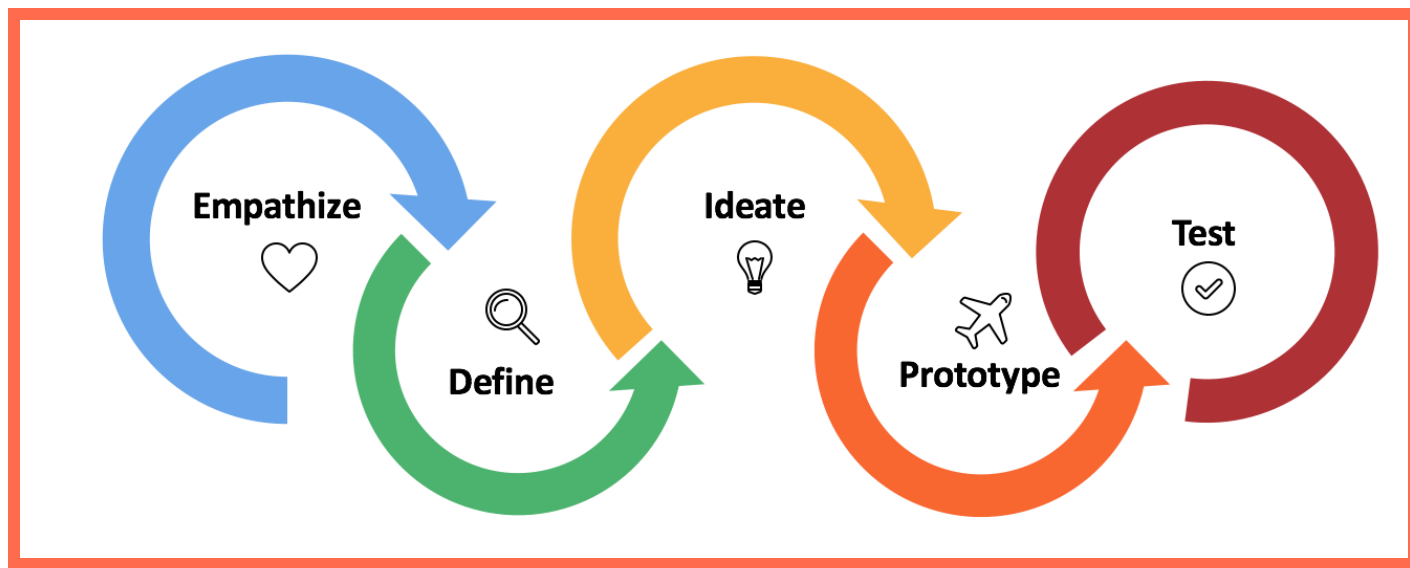


[Borrowed from spring2innovation.com](https://spring2innovation.com)

Rapid Prototyping in the post LLM era

RAPID PROTOTYPING OVERVIEW

AI prototyping today is not (always) about training new models from scratch—it's often about rapid iteration, user validation, and smart integration of LLMs into workflows



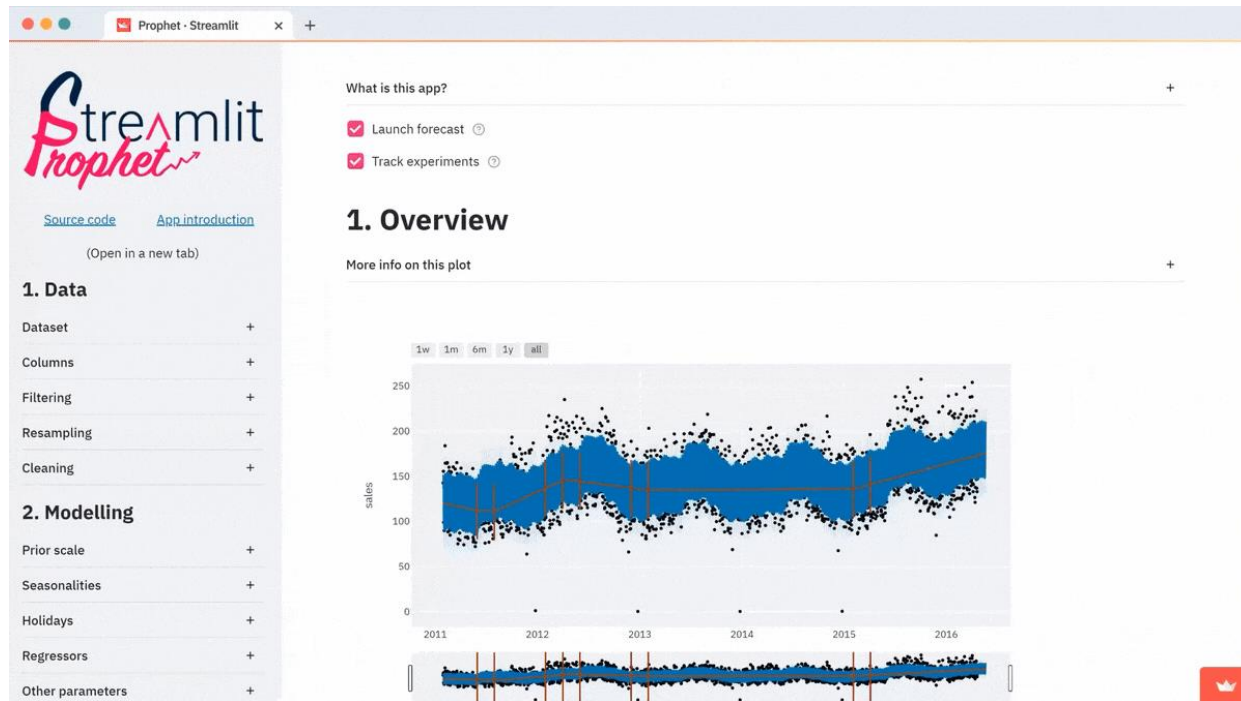
Make this go brr..

<https://spring2innovation.com/design-thinking-vs-user-centred-design/>

[Borrowed from spring2innovation.com](https://spring2innovation.com/)

WHAT IS A PROTOTYPE?

A prototype allows us to quickly AI features with real users before committing to full-scale development



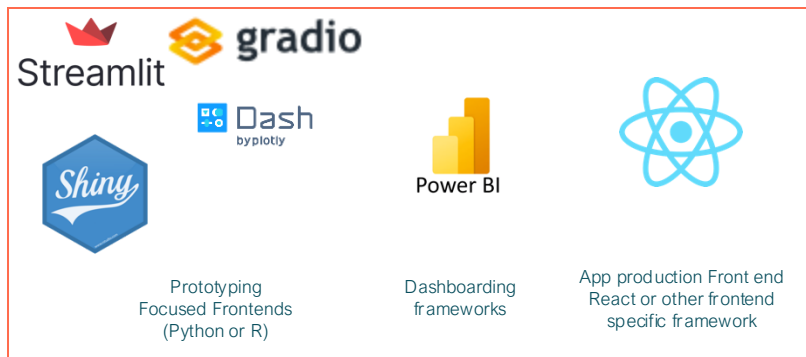
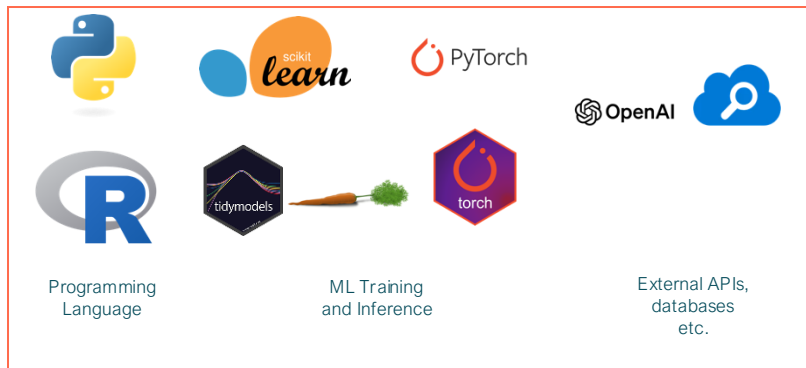
[Streamlit Prophet Time Series Forecasting Demo](#)



RAPID PROTOTYPING IN THE AI ERA

An AI model alone is not a product. A prototype must have a user interface that allows users to interact with AI predictions in a meaningful way

- Backend (what runs our main code):
 - Ingests data
 - Performs transformations and feature engineering
 - Runs model and produces output
 - Returns output in some relevant format
- Frontend (what the user interacts with):
 - A user might upload an excel file they want to run our model on
 - A user might set filters, and click a button to run the model
 - The user has some needs in terms of output visualization or information summary
 - The user makes some decision or performs an action based on the output



RAPID PROTOTYPING IN THE AI ERA

UI Matters! If the users can't envision what is possible, and how they can utilize the output, they can't give us the necessary feedback



Business
people when
shown a demo with
JSON-output
in terminal as UI



After
giving them
URL to a live
web app with
beautiful UI

Feedback $\approx$ Direction

Engagement $\approx$ Adoption

Understanding $\approx$
Expectation Management

Deployed Prototype =
Scaled (Individual) Testing

[Streamlit Prophet Time Series Forecasting Demo](#)

INTEGRATING PRE-TRAINED MODELS AND LLMS

The modern AI stack allows teams to build AI-powered applications without training models from scratch—reducing costs and speeding up development



LangChain



Pinecone

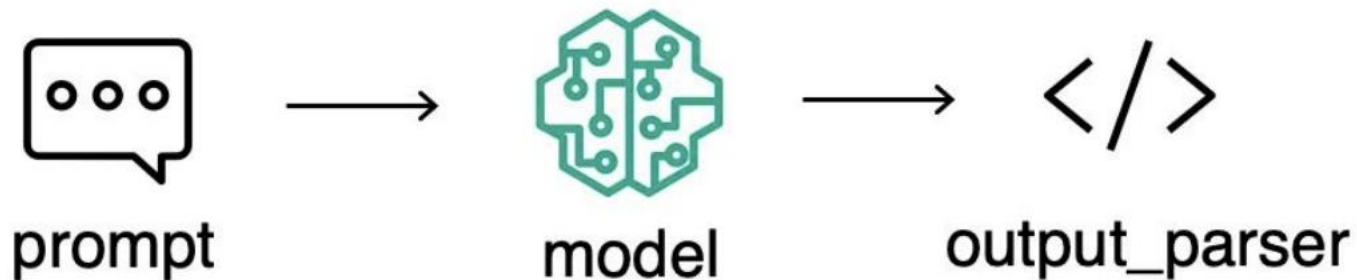


Hugging Face

- Example tools:
 - LLM API Calls (OpenAI)
 - Programmatically defined prompts
 - Tool-usage / Structured Outputs
 - Chunk+Embed
 - Semantic Search
 - Retrieval Augmented Generation (RAG)
 - AI Workflows
 - AI Agents
- Other pre-trained models



LLM takes a prompt as input, produces a response, and parses the output



PROGRAMMATICALLY CALLING LLMS

Most modern AI applications don't require training new models from scratch. Instead, we leverage cloud-based LLMs via APIs to power applications instantly

```
from poc import config
from openai import OpenAI

system_prompt = "You are a helpful assistant that can answer questions about the user's data."
prompt = "What is the capital of France?"
model = "gpt-4o"
client = OpenAI(
    api_key=config.AZURE_OPENAI_API_KEY.get_secret_value(),
    api_version=config.AZURE_OPENAI_VERSION,
    azure_endpoint=config.AZURE_OPENAI_BASE_URL,
)

response = client.chat.completions.create(
    model=model,
    messages=[
        {"role": "system", "content": system_prompt},
        {"role": "user", "content": prompt},
    ],
    temperature=0.2,
    max_tokens=1024,
)

print(response.choices[0].message.content)
```

WHAT IS A PROMPT?

We instruct LLMs to do our bidding by providing instructions and data/context through prompts

1. Description of task we want solved

2. Output examples

3. Specific instructions

You will now receive a project description in markdown format.

Your task is to summarize the scope of the project based on the information in the markdown text.

Here is an example of a project description:

«This competition concerns a design-and-build contract for constructing dedicated bus lanes, pedestrian and bicycle paths along National Road 22 between Hafslund and Dondern in the municipality of Sarpsborg. The project also includes the reconstruction of the roundabout at Dondern, the construction of two underpasses for pedestrians and cyclists, adjustments to stops, noise mitigation measures, and the demolition of several buildings. The goal of the project is to improve accessibility and traffic safety for both public transport and non-motorized road users along the route. The contract has a completion deadline of December 1, 2027.»

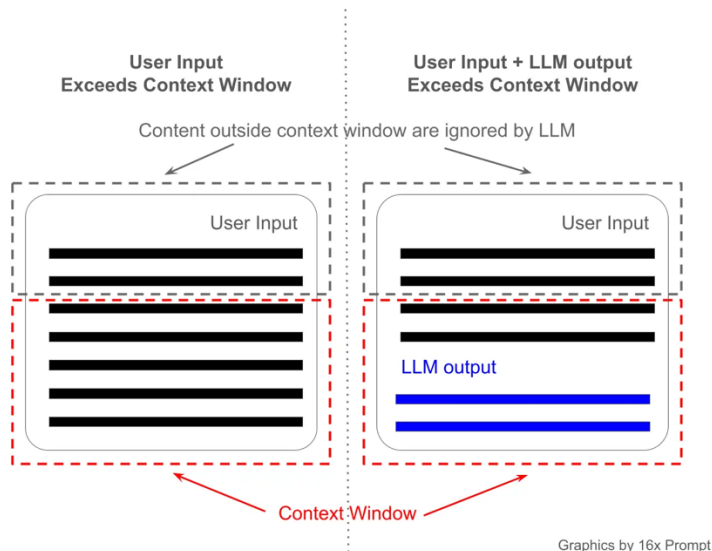
Important: Please note that you should write in Norwegian.

Base your work on the following content from the tender documentation in markdown format:

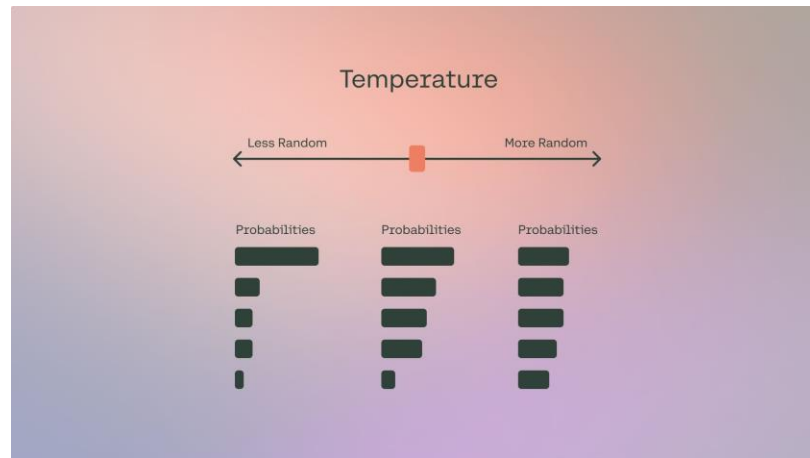
{markdown_text}

4. We can provide (long) context

Our input prompt must be within the context size of the model, and we can tweak the creativity of the model through temperature



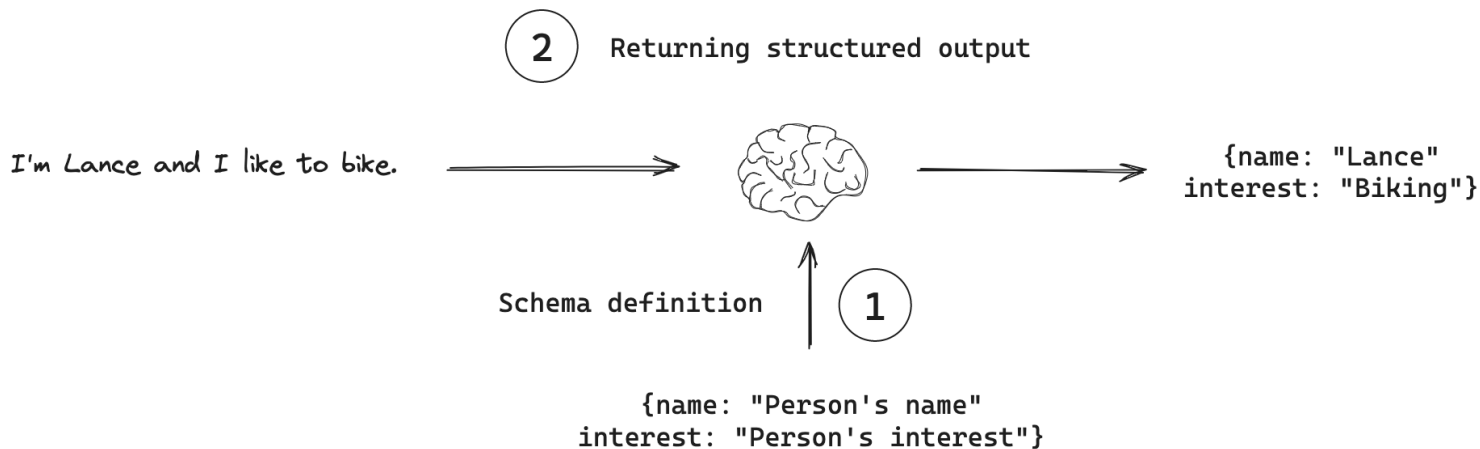
[Borrowed from Shlomi Boutnaru, Ph.D. @ Medium](#)



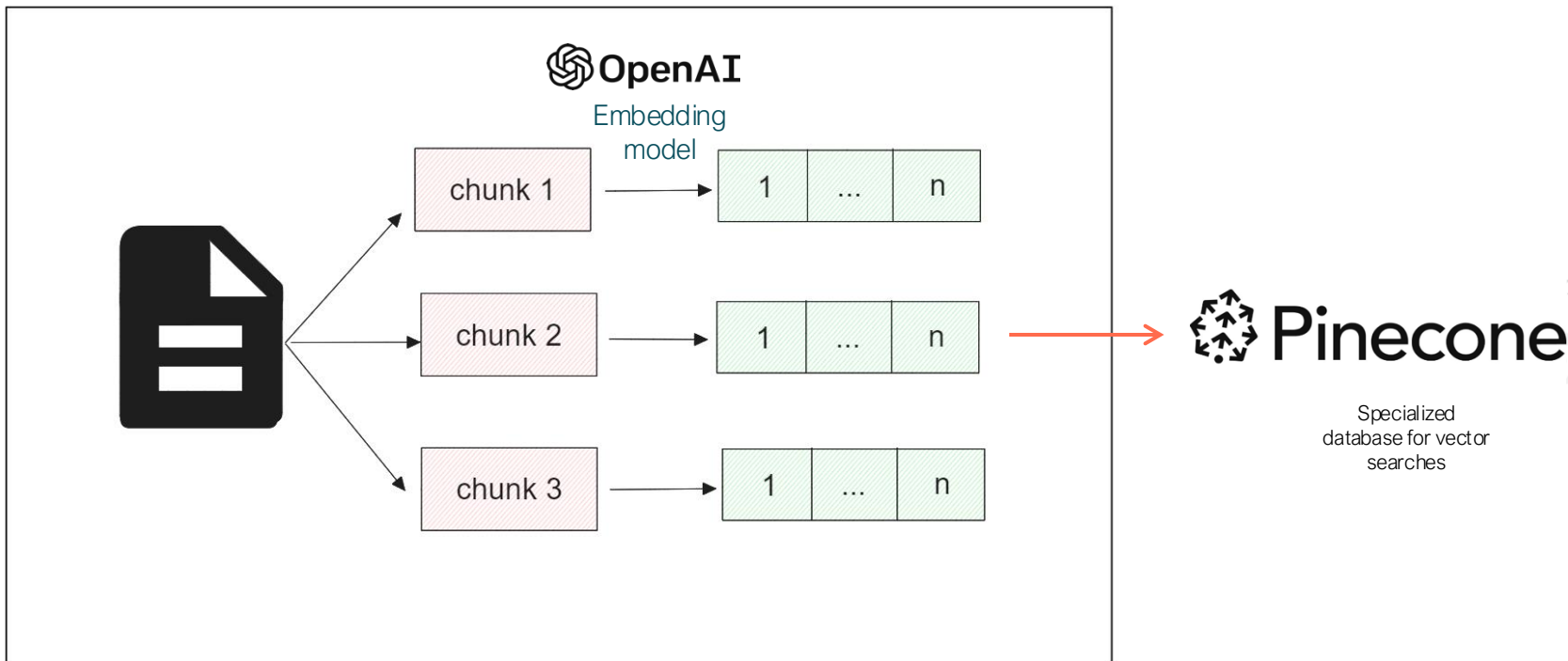
[Borrowed from cohere team @ Cohere](#)

AI TOOL-USE / STRUCTURED OUTPUTS

We can integrate tool-usage such as structured output to provide the model with capabilities such as output validation, web search etc.

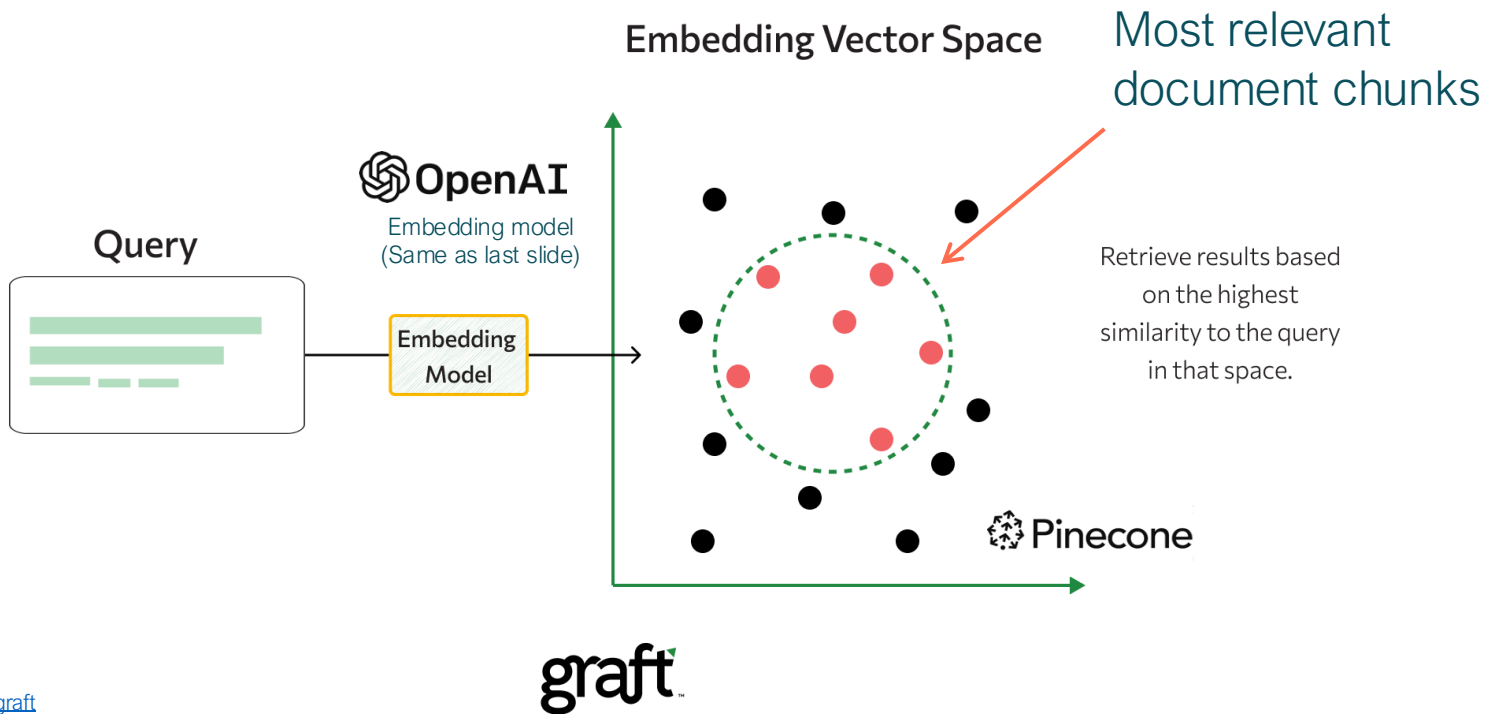


We can break long documents into smaller *chunks*, and utilize *embedding* models to encode their meaning in numeric vectors



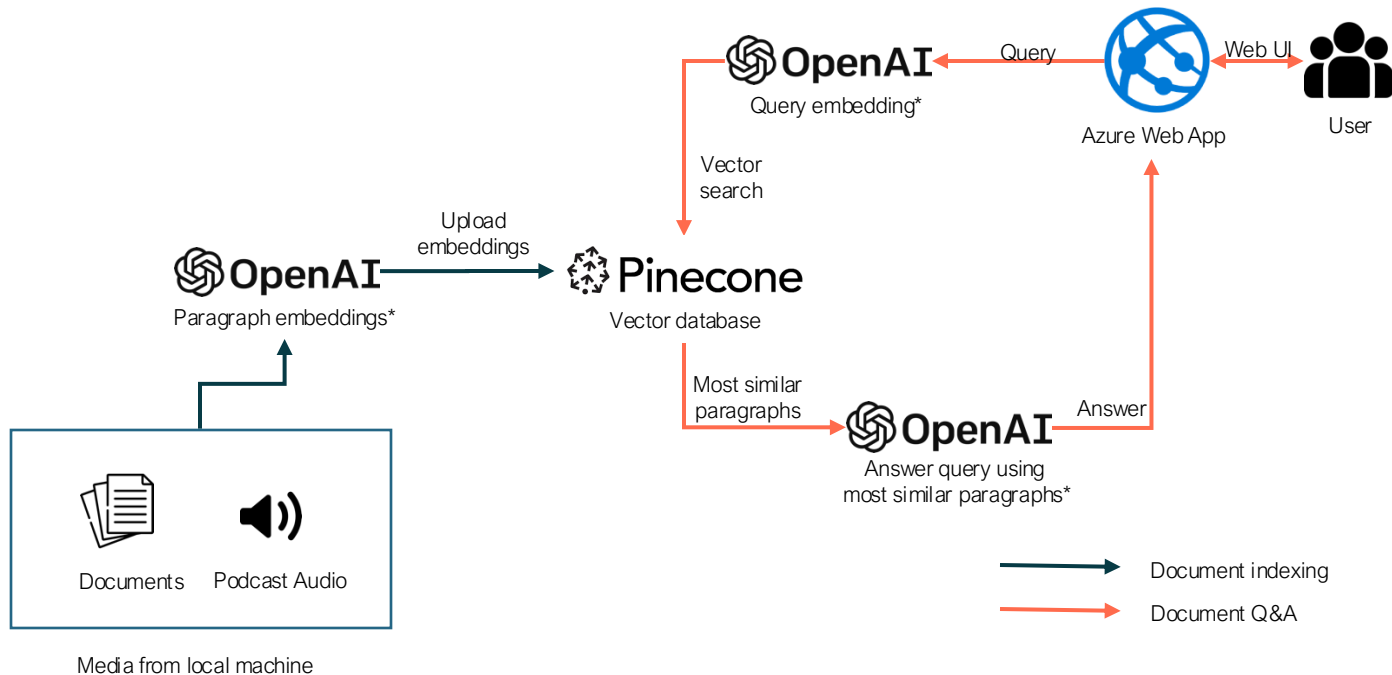
[Borrowed from Ruben Winastwan @ zilliz](#)

For a given input query, we can find which document chunks are most relevant through semantic search

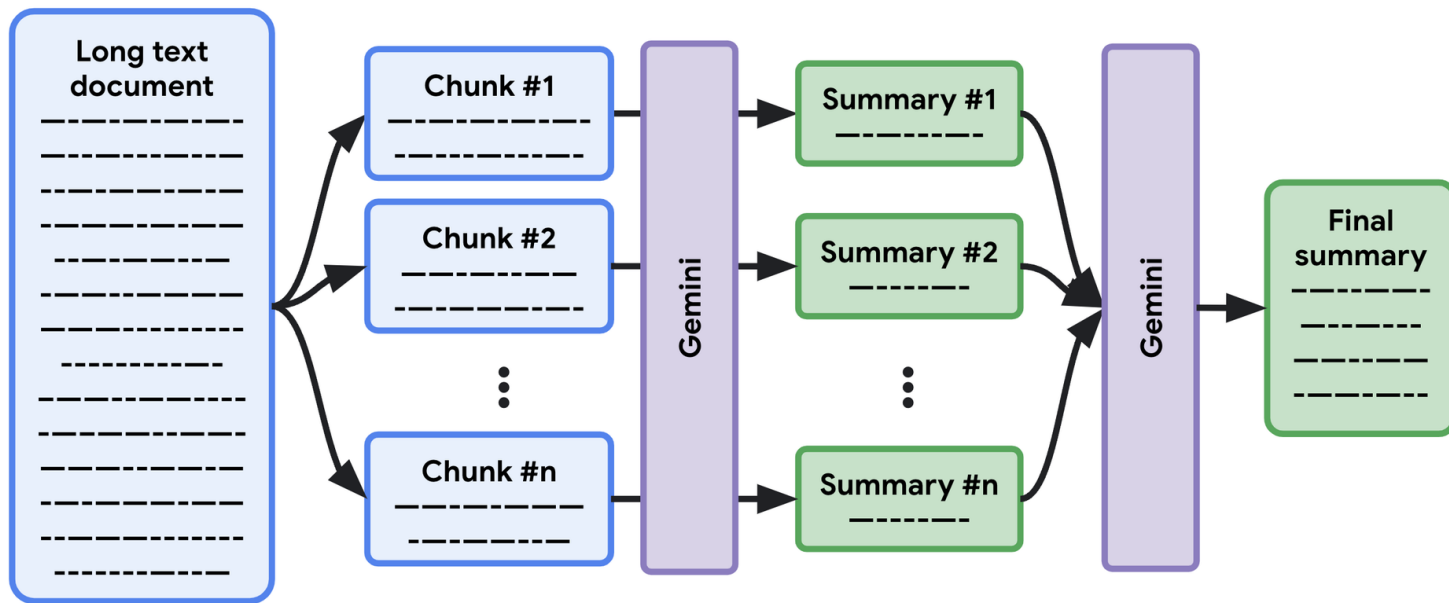


[Borrowed from JD Prater @graft](#)

Retrieval Augmented Generation performs a semantic search to find the most relevant document chunks to pass to the LLM as context



An AI Workflow sequentially runs multiple calls to the LLM in a pre-defined chain



[Borrowed Borrowed from google cloud - Long documentat summarization with Workflows and Gemini models](#)

An AI Agent is like an AI Workflow, but it uses an LLM to reason and plan what to do, and to decide when the task is complete



Case Study: AI Product for Company X

CASE INTRODUCTION

Company X works on large infrastructure projects that generate and rely on extensive documentation.

- Project teams must frequently **search, interpret, and summarize vast amounts of documents** to make critical decisions.

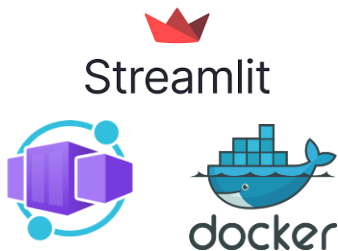
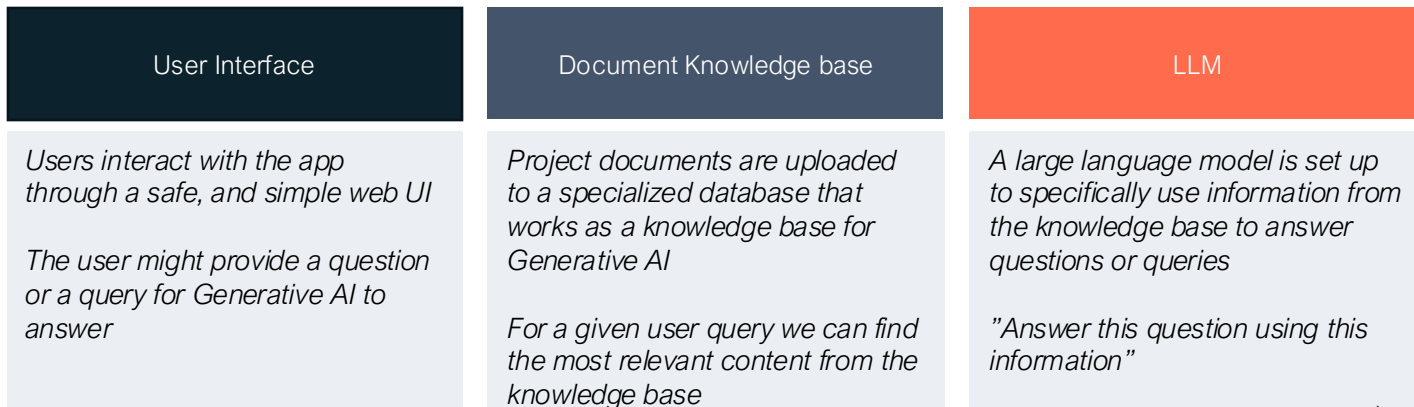
Case: Uncovering an Old Landfill

- During a **road construction project**, the team **discovered an old landfill** on-site. Who is responsible for the costs of removal and remediation?
- The answer is buried in **historical contracts, regulations, and project archives**



HIGH-LEVEL PROTOTYPE ARCHITECTURE

Together with project leads at Company X, we developed an AI Product for handling complex search and summarization across 100s of docs



Company X AI Tool: Rapid Retrieval of Critical Project Information Demo

SUMMARY OF AI TOOLKIT USAGE

The demo we just watched was built entirely with tools from the AI Toolkit we just discussed

- Offline pipeline running multiple pre-defined prompts, extracting key project information
- Offline pipeline applying structured output to extract contract demands in a structured format, and producing a dashboard over demand types/criticality
- Two stage hybrid RAG, finding and sorting relevant documents, and allowing user to select specific documents for deeper search, ensuring control over what information has been considered along the way
- Online, pre-defined prompts instructing on how to improve and generate different types of key documents, such as change messages

Wrap-up

LEARNING AND BUILDING WITH AI TOOLS

Use AI Tools! These tools democratize knowledge, empower you to build anything you imagine, and unblock you when you get stuck



Rapid problem-solving, learning,
design input, unblocking



Assist in code generation,
debugging, documenting



RESOURCES

Resources to dive deeper into ML/Deep Learning, LLMs, rapid prototyping, data science management and other topics

Theory

- Anything by [Andrej Karpathy](#)
- 3Blue1Brown [Neural Networks Series](#)
- Anything by [Weights and Biases](#)
- Anything by [Full Stack Deep Learning](#)

Coding

- [Python for R Users DataCamp Course](#)

ML Team and Product Management

- [Full Stack Deep Learning: ML Teams and Project Management](#)

AI Strategy:

- [Competing in the Age of AI \(Marco Iansiti, Karim R. Lakhani, 2020\)](#)

• Applied LLMs

- [OpenAI Dev Platform](#)
- [Langchain Quickstart](#)
- [Full Stack Deep Learning Series Launch an LLM App in an hour](#)
- [AI Agents Course Huggingface](#)

• Streamlit (UI)

- [Streamlit Quickstart](#)

Product Management :

- [Inspired: How to create tech products customers love](#)

Q&A



benjamin.aanes@crayon.no

